

Constrained sphere packing and the interpolation problem for positive definite functions

James Eldred Pascoe

Drexel University

Joint work with Sujit Sakharam Damase, IISc, Bangalore

A photograph of two young men standing in front of a large blackboard in a lecture hall. The man on the left is wearing a green t-shirt and a black jacket, while the man on the right is wearing a light-colored patterned sweater. The blackboard is filled with handwritten mathematical equations and notes in white chalk. The equations include various mathematical symbols, integrals, and algebraic expressions. The background shows a window with a wooden frame and a white wall.

Acknowledgments

We would like to thank:

- The Indian Institute of Science, Bangalore
- The Indian Statistical Institute, Bangalore
- J C Bose Fellowship JCB/2021/000041 of SERB
- Binational Science Foundation
- Fulbright Nehru Student and Scholar awards

for supporting this research.

Acknowledgments

We would like to thank:

- The Indian Institute of Science, Bangalore
- The Indian Statistical Institute, Bangalore
- J C Bose Fellowship JCB/2021/000041 of SERB
- Binational Science Foundation
- Fulbright Nehru Student and Scholar awards

for supporting this research.

Thanks to Jeff Remmel (1948-2017)
for teaching me representation theory.



Building international collaboration networks in under-served communities is key to a human role in further mathematical progress. It is easy to embrace roles in transient conflicts, the world is very high stress.

Sphere packing

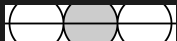
Sphere packing

Want to find the the optimal packing of spheres in some dimension.

Sphere packing

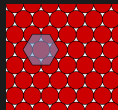
Want to find the optimal packing of spheres in some dimension.

$$n = 1$$



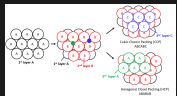
$$n = 2$$

Lagrange



$$n = 3$$

Hales (90s)



$$n = 8, 24$$

Viazovska et al. (2010s)

$$E_8,$$

Leech lattice

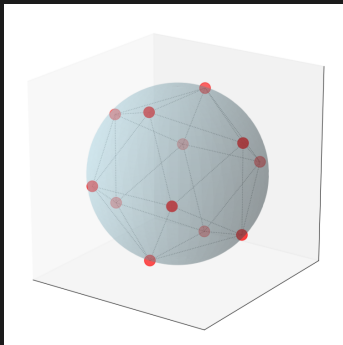
Spherical codes

Spherical codes

A **spherical code** is a set of points in the sphere in n real dimensional space. We say it has **minimum angle** to be the smallest θ such that every pair of distinct points in the code makes an angle of at least θ with respect to the origin.

Spherical codes

A **spherical code** is a set of points in the sphere in n real dimensional space. We say it has **minimum angle** to be the smallest θ such that every pair of distinct points in the code makes an angle of at least θ with respect to the origin.



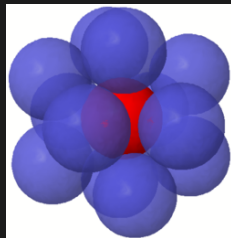
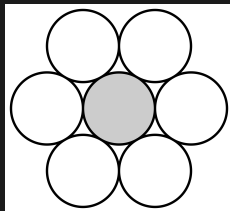
Kissing number problem

Kissing number problem

The **kissing number** for the sphere in n dimensions is the number of nonintersecting unit spheres that can be tangent to a single unit sphere.

Kissing number problem

The **kissing number** for the sphere in n dimensions is the number of nonintersecting unit spheres that can be tangent to a single unit sphere.



The kissing number is equal to the maximal size of a spherical code with minimum angle $\pi/3$.

Delsarte's method

Delsarte's method

Observation: Let $\pi/2 < \theta < \pi$. Any spherical code with minimum angle θ has cardinality less than

$$\frac{\cos \theta - 1}{\cos \theta}.$$

Delsarte's method

Observation: Let $\pi/2 < \theta < \pi$. Any spherical code with minimum angle θ has cardinality less than

$$\frac{\cos \theta - 1}{\cos \theta}.$$

Strategy: Given $n \in \mathbb{N}$ and $0 < \theta < \pi/2$ find the maximum $\tilde{\theta} > \pi/2$ such that there is a map from the sphere in n -dimensional space to the sphere in a larger dimensional space such that every spherical code with minimum angle θ is mapped to a code with minimum angle at least $\tilde{\theta}$.

Positive definite functions

Positive definite functions

Let G be group, let H be subgroup of G . A (completely) positive definite function

$$f : G/H \rightarrow \mathbb{R} \text{ (resp. self-adjoint operators)}$$

satisfies: $[f(g_i^* g_j)]_{i,j}$ is positive semidefinite for all $g_1, \dots, g_d \in G$.

Stinespring factorization

Stinespring factorization

Given a positive definite function $f : G/H \rightarrow \mathbb{R}$ it has a **Stinespring factorization**

$$f(g) = v^* \pi(g) v$$

where π is a representation of G and v is a vector which is invariant for $\pi|_H$. If f is continuous and G is a compact group, the Peter-Weyl decomposition gives this is a direct sum of irreducibles. (convex combination)

Stinespring factorization

Given a positive definite function $f : G/H \rightarrow \mathbb{R}$ it has a **Stinespring factorization**

$$f(g) = v^* \pi(g) v$$

where π is a representation of G and v is a vector which is invariant for $\pi|_H$. If f is continuous and G is a compact group, the Peter-Weyl decomposition gives this is a direct sum of irreducibles. (convex combination)

The homomorphic analogue of **Tomiyama's theorem** gives that f is in fact bi-invariant,

$$f(h_1 g h_2) = f(g), \forall h_1, h_2 \in H, g \in G.$$

Schoenberg's theorem on positive definite functions

Schoenberg's theorem on positive definite functions

Let $G = SO(n)$, $H = 1 \oplus SO(n-1)$, $S^{n-1} \cong G/H$. By bi-invariance we have that $f(\gamma)$ depends only on the $1-1$ entry of γ .

Schoenberg's theorem on positive definite functions

Let $G = SO(n)$, $H = 1 \oplus SO(n-1)$, $S^{n-1} \cong G/H$. By bi-invariance we have that $f(\gamma)$ depends only on the 1-1 entry of γ .

If f is positive definite and continuous, $f(g) = \sum_{i=1}^{\infty} a_i C_i^{(n)}(\gamma_{11})$ where $a_i \geq 0$ and the $C_i^{(n)}$ are Gegenbauer polynomials, a natural basis for bi-invariant functions. (We will normalize to have $C_i^{(n)}(1) = 1$) (Schoenberg '42, Bochner '41)

Operations on positive definite functions

Operations on positive definite functions

Given $C_i^{(n)}(\gamma_{11}) = v_i^* \pi_i(\gamma) v_i$, we see that

$$C_i^{(n)} C_j^{(n)} = \sum_{k=0}^{i+j} a_k C_k^{(n)}$$

where $\sum a_k = 1, a_k \geq 0$ as

$$C_i^{(n)}(\gamma_{11}) C_j^{(n)}(\gamma_{11}) = (v_i \otimes v_j)^* (\pi_i \otimes \pi_j)(\gamma) (v_i \otimes v_j)^*$$

and $\pi_i \otimes \pi_j$ may again be broken into irreducibles.

Kernel embedding

Kernel embedding

Theorem (Aronszajn's theorem)

Let X be a set. Let $A : X \times X \rightarrow \mathbb{C}$ such that $[A(x_i, x_j)]_{i,j}$ is positive semidefinite for arbitrary finite collections $x_1, \dots, x_d$.

There exists a Hilbert space $\mathcal{H}$ and $\iota_x : X \rightarrow \mathcal{H}$ such that $\langle \iota_x, \iota_y \rangle = A(x, y)$.

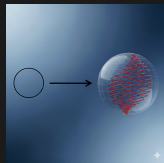
Kernel embedding

Theorem (Aronszajn's theorem)

Let X be a set. Let $A : X \times X \rightarrow \mathbb{C}$ such that $[A(x_i, x_j)]_{i,j}$ is positive semidefinite for arbitrary finite collections $x_1, \dots, x_d$.

There exists a Hilbert space $\mathcal{H}$ and $\iota_x : X \rightarrow \mathcal{H}$ such that $\langle \iota_x, \iota_y \rangle = A(x, y)$.

Thus a positive definite function $f : [-1, 1] \rightarrow \mathbb{R}$ such that $f(1) = 1$ induces an embedding via the kernel $f(\langle x, y \rangle)$ which lands in the sphere of the Hilbert space as $f(\langle x, x \rangle) = 1$.



Delsarte's method

Delsarte's method

Theorem (Delsarte '72)

Let $0 < \theta < \pi$. Let $g : [-1, 1] \rightarrow \mathbb{R}$ be a positive definite function on the sphere in n dimensions such that $g(1) = 1$ and $g(x) \leq 0$ if $x \leq \cos \theta$.

Let $\bar{g}$ the constant coefficient of g in its Gegenbauer expansion. (That is the average value of $g(\langle x, y \rangle)$ over x, y in the unit sphere. Let $\mathcal{C}$ be spherical code with minimal angle θ . Then,

$$|\mathcal{C}| \leq \frac{1}{\bar{g}}.$$

Delsarte's method

Theorem (Delsarte '72)

Let $0 < \theta < \pi$. Let $g : [-1, 1] \rightarrow \mathbb{R}$ be a positive definite function on the sphere in n dimensions such that $g(1) = 1$ and $g(x) \leq 0$ if $x \leq \cos \theta$.

Let $\bar{g}$ the constant coefficient of g in its Gegenbauer expansion. (That is the average value of $g(\langle x, y \rangle)$ over x, y in the unit sphere. Let $\mathcal{C}$ be spherical code with minimal angle θ . Then,

$$|\mathcal{C}| \leq \frac{1}{\bar{g}}.$$

The proof is essentially the kernel embedding trick plus the bound from earlier for the kernel given by $\frac{g-\bar{g}}{1-\bar{g}}$.

Delsarte's method

Theorem (Delsarte '72)

Let $0 < \theta < \pi$. Let $g : [-1, 1] \rightarrow \mathbb{R}$ be a positive definite function on the sphere in n dimensions such that $g(1) = 1$ and $g(x) \leq 0$ if $x \leq \cos \theta$.

Let $\bar{g}$ the constant coefficient of g in its Gegenbauer expansion. (That is the average value of $g(\langle x, y \rangle)$ over x, y in the unit sphere. Let $\mathcal{C}$ be spherical code with minimal angle θ . Then,

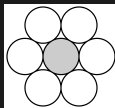
$$|\mathcal{C}| \leq \frac{1}{\bar{g}}.$$

The proof is essentially the kernel embedding trick plus the bound from earlier for the kernel given by $\frac{g-\bar{g}}{1-\bar{g}}$.

Linear programming methods: Delsarte-Goethals-Seidel, '77 Kabitsansky-Levenshtein '78.

Kissing number in $\mathbb{R}^2$

Kissing number in $\mathbb{R}^2$



The polynomial

$$p(t) = \frac{4}{9}(t - 1/2)(t + 1/2)^2(1 + t)$$

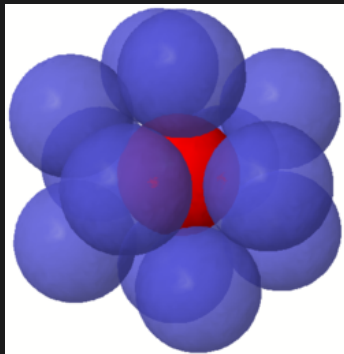
satisfies

$$p(x) = \frac{1}{6}C_0^{(2)}(x) + \frac{1}{3}C_1^{(2)}(x) + \frac{5}{18}C_2^{(2)}(x) + \frac{1}{6}C_3^{(2)}(x) + \frac{1}{18}C_4^{(2)}(x)$$

$$\text{i.e. } p(\cos \theta) = \frac{1}{6} + \frac{1}{3} \cos \theta + \frac{5}{18} \cos 2\theta + \frac{1}{6} \cos 3\theta + \frac{1}{18} \cos 4\theta$$

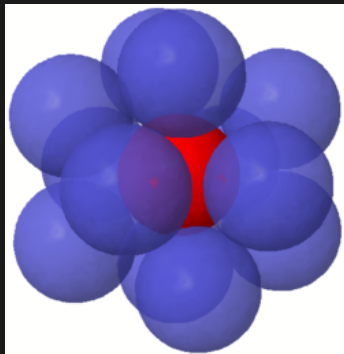
Kissing number in $\mathbb{R}^3$

Kissing number in $\mathbb{R}^3$



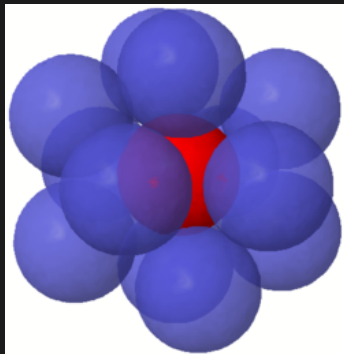
The kissing number is 12, but Delsarte gives an estimate greater than 13.

Kissing number in $\mathbb{R}^3$



The kissing number is 12, but Delsarte gives an estimate greater than 13. Packing is loose.

Kissing number in $\mathbb{R}^3$



The kissing number is 12, but Delsarte gives an estimate greater than 13. Packing is loose. Musin resolves using two kernel estimates. ('06)

Delsarte sharpness

Delsarte sharpness

Let $X \subseteq [-1, 1]$.

Delsarte sharpness

Let $X \subseteq [-1, 1]$. We say X is **Delsarte sharp** if there exists a spherical code $\mathcal{C}$ and a $g : [-1, 1] \rightarrow \mathbb{R}$ be a positive definite function on the sphere in n dimensions such that $g(1) = 1$ and $g(x) \leq 0$ if $x \in X$ such that

$$|\mathcal{C}| = \frac{1}{\bar{g}}$$

where $\bar{g}$ is the spherical average of g .

Delsarte sharpness

Let $X \subseteq [-1, 1]$. We say X is **Delsarte sharp** if there exists a spherical code $\mathcal{C}$ and a $g : [-1, 1] \rightarrow \mathbb{R}$ be a positive definite function on the sphere in n dimensions such that $g(1) = 1$ and $g(x) \leq 0$ if $x \in X$ such that

$$|\mathcal{C}| = \frac{1}{\bar{g}}$$

where $\bar{g}$ is the spherical average of g .

Sharp

$n = 2$ kissing number

$n = 8, 24$ kissing number

Sloane, Odlyzko, 79

Not Sharp

$n = 3, 4$ kissing number

Musin

$X = \{-1, 1/2, -1/2, 1\}, n = 10$

Damase, P. ($\frac{1}{\bar{g}} = 46.$)

Cohn-Elkies method

Cohn-Elkies method

Theorem (Cohn-Elkies, '03)

Suppose $f : \mathbb{R}^n \rightarrow \mathbb{R}$ is an admissible function, is not identically zero, and satisfies the following two conditions:

- 1 $f(x) \leq 0$ for $|x| \geq 1$, and
- 2 $\hat{f}(t) \geq 0$ for all t .

Then the center density of n -dimensional sphere packings is bounded above by

$$\frac{f(0)}{2^n \hat{f}(0)}.$$

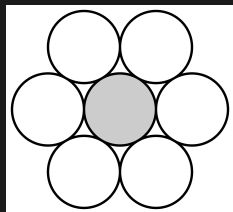
Constrained angle codes

Constrained angle codes

We say a spherical code $\mathcal{C}$ has cosines in X if the angles between distinct points have cosines contained in X .

Constrained angle codes

We say a spherical code $\mathcal{C}$ has cosines in X if the angles between distinct points have cosines contained in X .



For example, the spherical code given by six equispaced points on the circle has cosines in $\{0, -1, 1/2, -1/2\}$.

Interpolation for positive definite functions

Interpolation for positive definite functions

Let X be a finite subset of $[-1, 1]$. Let $f : X \rightarrow \mathbb{R}$.

Interpolation for positive definite functions

Let X be a finite subset of $[-1, 1]$. Let $f : X \rightarrow \mathbb{R}$.

When is there a positive definite function on the sphere in n -dimensions $\tilde{f} : [-1, 1] \rightarrow \mathbb{R}$ such that $\tilde{f}|_X = f$?

Partially defined positive definite functions

Partially defined positive definite functions

Let G be group, let H be subgroup of G . Let $\Gamma \subseteq H \backslash G / H$ such that $\Gamma = \Gamma^{-1}$, $HeH \in \Gamma$.

Partially defined positive definite functions

Let G be group, let H be subgroup of G . Let $\Gamma \subseteq H \backslash G / H$ such that $\Gamma = \Gamma^{-1}$, $HeH \in \Gamma$. A (completely) positive definite function

$$f : \Gamma \rightarrow \mathbb{R} \text{ (resp. self-adjoint operators)}$$

is a map which naturally induces a completely positive map on the operator system corresponding to Γ inside the group algebra $\mathbb{C}[G]$.

Partially defined positive definite functions

Let G be group, let H be subgroup of G . Let $\Gamma \subseteq H \backslash G / H$ such that $\Gamma = \Gamma^{-1}$, $HeH \in \Gamma$. A (completely) positive definite function

$$f : \Gamma \rightarrow \mathbb{R} \text{ (resp. self-adjoint operators)}$$

is a map which naturally induces a completely positive map on the operator system corresponding to Γ inside the group algebra $\mathbb{C}[G]$.
c.f. Krein, Jorgensen, etc. on similar but different interpolation problems.

Arveson extension and pathological Class I representations

Arveson extension and pathological Class I representations

- One uses the Arveson extension theorem to obtain an extension to the entire domain.

Arveson extension and pathological Class I representations

- One uses the Arveson extension theorem to obtain an extension to the entire domain.
- However, as one is extending to the entire (discrete) group algebra, one may obtain discontinuous representations of $SO(n)$.

Arveson extension and pathological Class I representations

- One uses the Arveson extension theorem to obtain an extension to the entire domain.
- However, as one is extending to the entire (discrete) group algebra, one may obtain discontinuous representations of $SO(n)$.

Say G/H is **pathological** if there exists a positive definite function on G/H which is not a pointwise limit of continuous ones.

Arveson extension and pathological Class I representations

- One uses the Arveson extension theorem to obtain an extension to the entire domain.
- However, as one is extending to the entire (discrete) group algebra, one may obtain discontinuous representations of $SO(n)$.

Say G/H is **pathological** if there exists a positive definite function on G/H which is not a pointwise limit of continuous ones.

Theorem (Damase-P.)

$SO(\infty)/1 \oplus SO(\infty)$ and $SO(2)/\{1\}$ are not pathological.

Continuous extension problem

Continuous extension problem

Let X be a finite subset of $[-1, 1]$. Let $f : X \rightarrow \mathbb{R}$. When is there a continuous positive definite function on the sphere in n -dimensions $\tilde{f} : [-1, 1] \rightarrow \mathbb{R}$ such that $\tilde{f}|_X = f$?

Continuous extension problem

Let X be a finite subset of $[-1, 1]$. Let $f : X \rightarrow \mathbb{R}$. When is there a continuous positive definite function on the sphere in n -dimensions $\tilde{f} : [-1, 1] \rightarrow \mathbb{R}$ such that $\tilde{f}|_X = f$?

Theorem (Damase, P.)

Let f correspond to a solvable continuous extension problem. There is a polynomial solution $\tilde{f}$.

Polynomial Delsarte sharpness

Polynomial Delsarte sharpness

Theorem (Nonantipodal sharpness)

Let $-1 \notin X$. If X is Delsarte sharp, then there exists a positive definite polynomial g with $\frac{1}{g} = |C|$.

Polynomial Delsarte sharpness

Theorem (Nonantipodal sharpness)

Let $-1 \notin X$. If X is Delsarte sharp, then there exists a positive definite polynomial g with $\frac{1}{g} = |C|$.

Theorem (Projective sharpness)

If $X = -X$ is Delsarte sharp, then any even function h such that $h - 2\bar{g}$ is positive definite, $h|_{X \setminus \{\pm 1\}} \leq 0$ and $h(1) = 1$ is a polynomial.

Polynomial Delsarte sharpness

Theorem (Nonantipodal sharpness)

Let $-1 \notin X$. If X is Delsarte sharp, then there exists a positive definite polynomial g with $\frac{1}{g} = |C|$.

Theorem (Projective sharpness)

If $X = -X$ is Delsarte sharp, then any even function h such that $h - 2\bar{g}$ is positive definite, $h|_{X \setminus \{\pm 1\}} \leq 0$ and $h(1) = 1$ is a polynomial.

Viazovska et al. functions for Cohn-Elkies are modular functions. (Low algebraic complexity.)

Delsarte hallucinations

Delsarte hallucinations

- Consider $X = \{-1, 1/2, -1/2, 1\}$, $n = 10$. There is no corresponding constrained angle code, but we have $\frac{1}{g} = 46$ is an integer. Many such examples exist of **Delsarte hallucinations**.

Delsarte hallucinations

- Consider $X = \{-1, 1/2, -1/2, 1\}$, $n = 10$. There is no corresponding constrained angle code, but we have $\frac{1}{g} = 46$ is an integer. Many such examples exist of **Delsarte hallucinations**.
- On the other hand, the example $X = \{1/3, -1/3, 1\}$ has a spherical code with 10 points in the 5 dimensional sphere, but the sign pattern comes from a Petersen graph.

Delsarte hallucinations

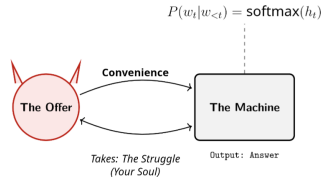
- Consider $X = \{-1, 1/2, -1/2, 1\}$, $n = 10$. There is no corresponding constrained angle code, but we have $\frac{1}{g} = 46$ is an integer. Many such examples exist of **Delsarte hallucinations**.
- On the other hand, the example $X = \{1/3, -1/3, 1\}$ has a spherical code with 10 points in the 5 dimensional sphere, but the sign pattern comes from a Petersen graph.
- Given an X , determining if there exists a subset of the n -sphere with cosines in X of some size k is a difficult combinatorial matrix completion problem.

The future is now

Algebra is the offer made by the devil to the mathematician. The devil says: I will give you this powerful machine, it will answer any question you like. All you need to do is give me your soul: give up geometry and you will have this marvelous machine.



—Sir Michael Atiyah, 2002

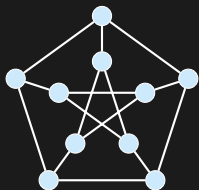


AI is the offer made by the devil to the mind.

The devil says: I will give you this powerful machine, it will complete any thought you like.

All you need to do is give me your soul: give up **the struggle** and you will have this marvelous machine.

— Gemini, 2025



The Petersen Graph

Seidel Matrix S

$$\begin{pmatrix} 0 & - & - & 1 & 1 & - & 1 & 1 & 1 & 1 \\ - & 0 & - & - & 1 & 1 & - & 1 & 1 & 1 \\ - & - & 0 & - & - & 1 & 1 & - & 1 & 1 \\ 1 & - & - & 0 & - & 1 & 1 & 1 & - & 1 \\ 1 & 1 & - & - & 0 & 1 & 1 & 1 & 1 & - \\ - & 1 & 1 & 1 & 1 & 0 & 1 & - & 1 & - \\ 1 & - & 1 & 1 & 1 & 1 & 0 & 1 & - & - \\ 1 & 1 & - & 1 & 1 & - & 1 & 0 & - & - \\ 1 & 1 & 1 & - & 1 & 1 & - & - & 0 & - \\ 1 & 1 & 1 & 1 & - & - & - & - & - & 0 \end{pmatrix}$$

Spectrum: $3^{(5)}, -3^{(5)}$

Gram Matrix G

Construct $X = \{1, \pm 1/3\}$:

$$G = I + \frac{1}{3}S \succeq 0$$

Discovery: AI = 爱

- Example suggested by **Gemini**.
- Verified & Formalized by **Damase & P.**

Soft Thresholding

Soft Thresholding

Given an empirical correlation matrix of rank N , (low sample statistics) one may want to threshold small entries to 0 to obtain sparsity.

However, such a process does not preserve the positive definiteness, as it is not a positive definite function.

Soft Thresholding

Given an empirical correlation matrix of rank N , (low sample statistics) one may want to threshold small entries to 0 to obtain sparsity.

However, such a process does not preserve the positive definiteness, as it is not a positive definite function.

The natural measure of *faithfulness* is the maximum constant coefficient in the Gegenbauer expansion of a positive definite function vanishing on some set K to be thresholded to 0. (Closer, to one, closer to linear, and hence easier to recover the original data.)

Soft Thresholding

Given an empirical correlation matrix of rank N , (low sample statistics) one may want to threshold small entries to 0 to obtain sparsity.

However, such a process does not preserve the positive definiteness, as it is not a positive definite function.

The natural measure of *faithfulness* is the maximum constant coefficient in the Gegenbauer expansion of a positive definite function vanishing on some set K to be thresholded to 0. (Closer, to one, closer to linear, and hence easier to recover the original data.)

Theorem (Damase, P.)

The faithfulness constant of $[-\varepsilon, \varepsilon]$ does not go to 1 if $N \geq 4$.

Soft Thresholding

Soft Thresholding

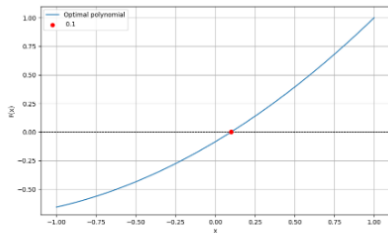


FIGURE 1. The thresholding function for $K = \{0.1\}, n = 3$

